# DataTransChall

AUTHOR
Vaishnavi Thorat

## # Data Translation Project

```
# Install packages if missing
if (!requireNamespace("ggplot2", quietly = TRUE)) install.packages("ggplot2")
if (!requireNamespace("dplyr", quietly = TRUE)) install.packages("dplyr")
if (!requireNamespace("fixest", quietly = TRUE)) install.packages("fixest")
if (!requireNamespace("modelsummary", quietly = TRUE)) install.packages("modelsummary")

library(ggplot2)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(fixest)
library(modelsummary)
```

```
load("DWI_Data.rdata")

df <- dwi   # IMPORTANT: set df as the dataset
names(df)   # check columns
```

```
 [1] "Date"       "Alcohol1"   "Alcohol2"   "male"       "white"
 [6] "recidivism" "acc"        "aged"       "year"       "bac1"
[11] "bac2"
```

```
df <- df |>
  dplyr::mutate(
    bac = as.numeric(bac1),
    recidivism = as.numeric(recidivism),
    male = as.numeric(male),
    white = as.numeric(white),
    aged = as.numeric(aged),
    acc = as.numeric(acc)
```

```
  ) |>
  dplyr::filter(!is.na(bac), !is.na(recidivism))
```

```
# Debug checks
class(df)
```

```
[1] "data.table" "data.frame"
```

```
dim(df)
```

```
[1] 214558     12
```

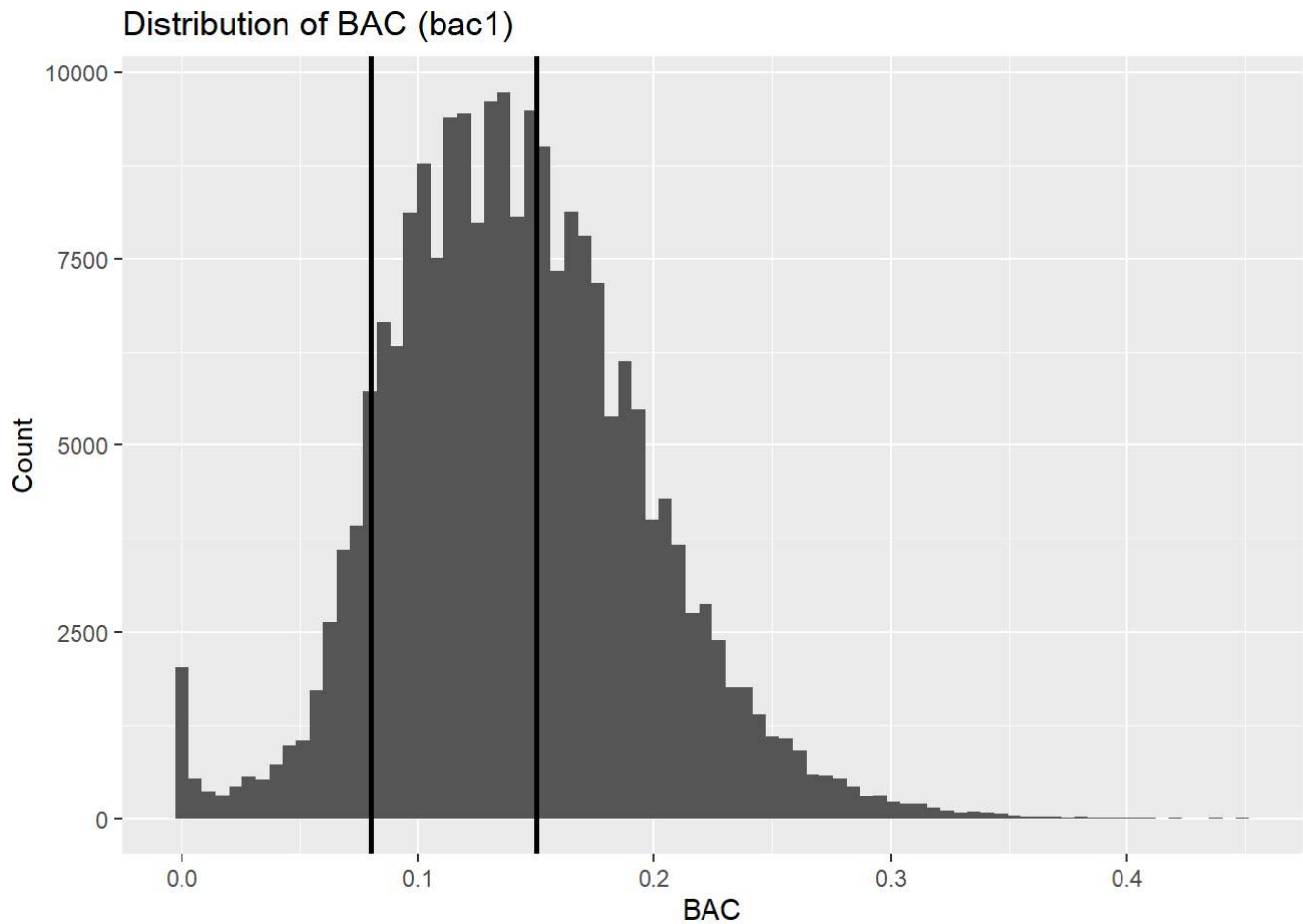
```
"recidivism" %in% names(df)
```

```
[1] TRUE
```

```
names(df)
```

```
 [1] "Date"       "Alcohol1"   "Alcohol2"   "male"       "white"
 [6] "recidivism" "acc"        "aged"       "year"       "bac1"
[11] "bac2"       "bac"
```

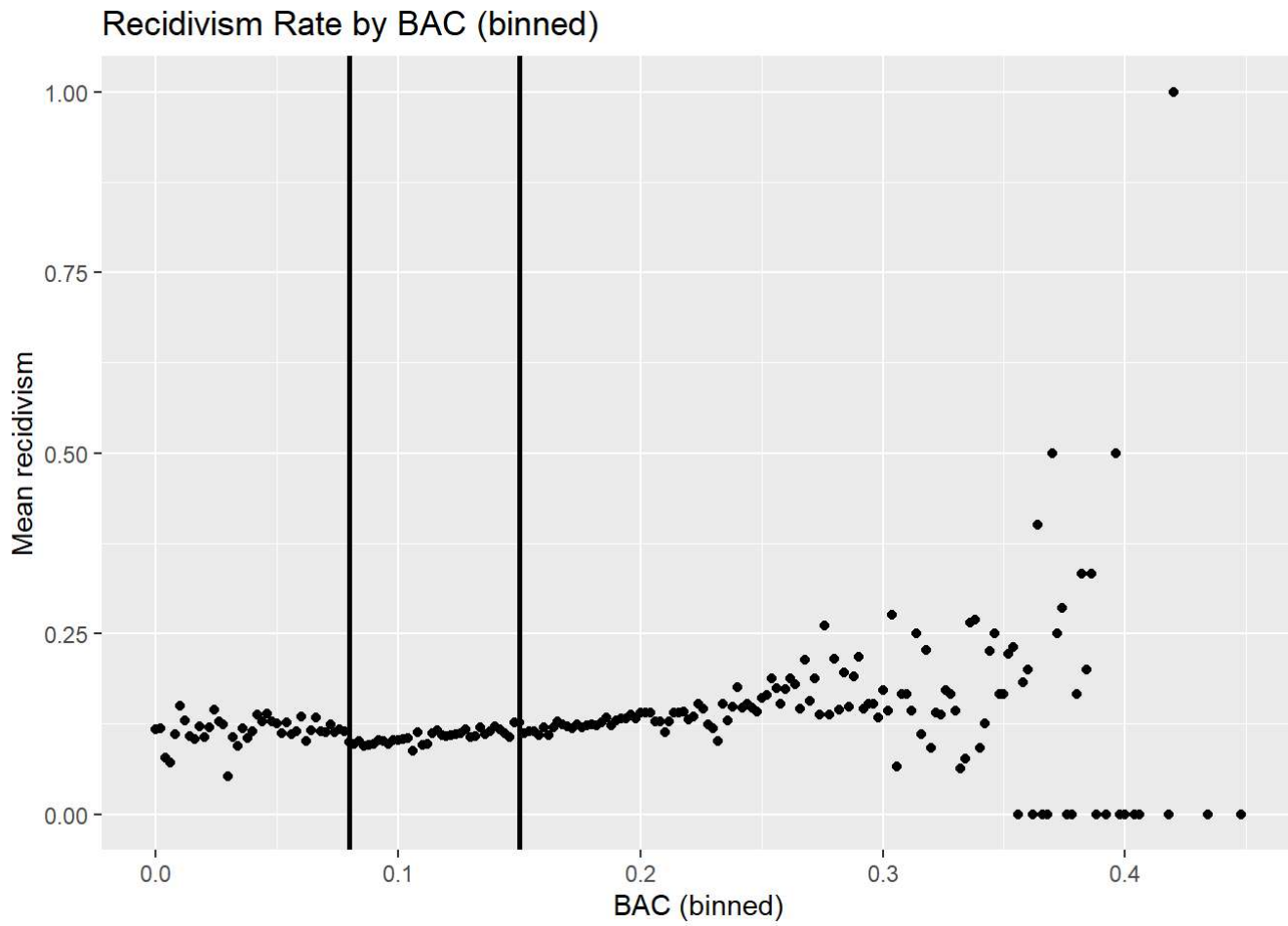
```
ggplot(df, aes(x = bac)) +
  geom_histogram(bins = 80) +
  geom_vline(xintercept = 0.08, linewidth = 1) +
  geom_vline(xintercept = 0.15, linewidth = 1) +
  labs(title = "Distribution of BAC (bac1)", x = "BAC", y = "Count")
```




This figure shows the distribution of BAC levels in the dataset. The vertical lines represent the legal thresholds of 0.08 and 0.15 BAC, which correspond to DUI penalties used in the regression discontinuity design.

```
df_binned <- df %>%
  mutate(bac_bin = floor(bac / 0.002) * 0.002) %>%
  group_by(bac_bin) %>%
  summarise(recid_rate = mean(recidivism), n = n(), .groups = "drop")

ggplot(df_binned, aes(x = bac_bin, y = recid_rate)) +
  geom_point() +
  geom_vline(xintercept = 0.08, linewidth = 1) +
  geom_vline(xintercept = 0.15, linewidth = 1) +
  labs(title = "Recidivism Rate by BAC (binned)", x = "BAC (binned)", y = "Mean recidivism")
```




This figure shows the relationship between BAC and the average recidivism rate. BAC values are grouped into small bins to makethe pattern easier to observe. The vertical lines indicate the legal BAC thresholds.

```
bw <- 0.05

df08 <- df %>%
  mutate(
    treat = ifelse(bac >= 0.08, 1, 0),
    x = bac - 0.08
  ) %>%
  filter(abs(x) <= bw)

m08 <- feols(
  recidivism ~ treat + x + treat:x + male + white + aged + acc,
  data = df08,
  vcov = "HC1"
)

summary(m08)
```

```
OLS estimation, Dep. Var.: recidivism
Observations: 89,967
Standard-errors: Heteroskedasticity-robust
             Estimate Std. Error    t value   Pr(>|t|)
```

```
(Intercept)  0.105923   0.005355  19.780462  < 2.2e-16 ***
treat       -0.024019   0.004353  -5.518009 3.4380e-08 ***
x           -0.042868   0.186632  -0.229691 8.1833e-01
male         0.033180   0.002329  14.247341  < 2.2e-16 ***
white        0.016224   0.002802   5.789186 7.0962e-09 ***
aged        -0.000854   0.000085 -10.047433  < 2.2e-16 ***
acc          0.004210   0.003452   1.219481 2.2266e-01
treat:x      0.438090   0.203798   2.149631 3.1587e-02 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.308472   Adj. R2: 0.003566
```

To estimate the causal effect of crossing the BAC threshold, we estimate a local linear regression discontinuity model around the cutoff. The treatment variable equals 1 when BAC is above the legal limit.

```r
df15 <- df %>%
  mutate(
    treat = ifelse(bac >= 0.15, 1, 0),
    x = bac - 0.15
  ) %>%
  filter(abs(x) <= bw)

m15 <- feols(
  recidivism ~ treat + x + treat:x + male + white + aged + acc,
  data = df15,
  vcov = "HC1"
)

summary(m15)
```

```
OLS estimation, Dep. Var.: recidivism
Observations: 138,925
Standard-errors: Heteroskedasticity-robust
             Estimate Std. Error    t value   Pr(>|t|)
(Intercept)  0.119422   0.004222  28.282619  < 2.2e-16 ***
treat       -0.006600   0.003336  -1.978636 4.7859e-02 *
x            0.406367   0.077838   5.220666 1.7854e-07 ***
male         0.028511   0.001972  14.460668  < 2.2e-16 ***
white        0.005137   0.002480   2.071589 3.8306e-02 *
aged        -0.000740   0.000073 -10.192076  < 2.2e-16 ***
acc         -0.007502   0.002423  -3.096137 1.9610e-03 **
treat:x      0.017119   0.122792   0.139413 8.8912e-01
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.318869   Adj. R2: 0.00268
```

```r
modelsummary(
  list("RDD cutoff 0.08" = m08, "RDD cutoff 0.15" = m15),
  statistic = "({std.error})",
  stars = TRUE,
```

```
  output = "markdown"
)
```

| | RDD cutoff 0.08 | RDD cutoff 0.15 |
|---|---|---|
| (Intercept) | 0.106*** | 0.119*** |
| | (0.005) | (0.004) |
| treat | -0.024*** | -0.007* |
| | (0.004) | (0.003) |
| x | -0.043 | 0.406*** |
| | (0.187) | (0.078) |
| male | 0.033*** | 0.029*** |
| | (0.002) | (0.002) |
| white | 0.016*** | 0.005* |
| | (0.003) | (0.002) |
| aged | -0.001*** | -0.001*** |
| | (0.000) | (0.000) |
| acc | 0.004 | -0.008** |
| | (0.003) | (0.002) |
| treat × x | 0.438* | 0.017 |
| | (0.204) | (0.123) |
| Num.Obs. | 89967 | 138925 |
| R2 | 0.004 | 0.003 |
| R2 Adj. | 0.004 | 0.003 |
| AIC | 43706.8 | 76692.5 |
| BIC | 43782.1 | 76771.2 |
| RMSE | 0.31 | 0.32 |
| Std.Errors | Heteroskedasticity-robust | Heteroskedasticity-robust |

- p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

---

```
balance_models_08 <- list(
  male = feols(male ~ treat + x + treat:x, data = df08, vcov = "HC1"),
  white = feols(white ~ treat + x + treat:x, data = df08, vcov = "HC1"),
  aged = feols(aged ~ treat + x + treat:x, data = df08, vcov = "HC1"),
  acc = feols(acc ~ treat + x + treat:x, data = df08, vcov = "HC1")
)

modelsummary(balance_models_08, statistic = "({std.error})", stars = TRUE, output = "markdown")
```

| | male | white | aged | acc |
|---|---|---|---|---|
| (Intercept) | 0.784*** | 0.846*** | 33.920*** | 0.083*** |
| | (0.005) | (0.004) | (0.135) | (0.003) |
| treat | 0.006 | 0.006 | -0.140 | -0.003 |
| | (0.006) | (0.005) | (0.164) | (0.004) |
| x | -0.210 | 0.079 | -69.164*** | -1.096*** |
| | (0.240) | (0.214) | (7.220) | (0.186) |
| treat × x | 0.307 | 0.016 | 76.049*** | 1.888*** |
| | (0.263) | (0.234) | (7.844) | (0.203) |
| Num.Obs. | 89967 | 89967 | 89967 | 89967 |
| R2 | 0.000 | 0.000 | 0.002 | 0.002 |
| R2 Adj. | 0.000 | 0.000 | 0.002 | 0.001 |
| AIC | 93116.8 | 68729.8 | 695916.9 | 39508.2 |
| BIC | 93154.5 | 68767.4 | 695954.5 | 39545.9 |
| RMSE | 0.41 | 0.35 | 11.57 | 0.30 |
| Std.Errors | Heteroskedasticity-robust | Heteroskedasticity-robust | Heteroskedasticity-robust | Heteroskedasticity-robust |

- p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

## Conclusion:

The analysis shows that crossing the legal BAC threshold affects future driving behavior. Drivers with BAC levels just above 0.08 have a lower probability of recidivism compared to those just below the cutoff, suggesting that DUI penalties may deter repeat offenses. The effect at the 0.15 threshold is smaller but still present. Overall, the results indicate that stricter DUI enforcement can reduce repeat drunk-driving behavior.